17	A sound wave of frequency 400 Hz is travelling in air at a speed of 320 m s^{-1}. What is the difference in phase between two points on the wave 0.2 m apart in the direction of travel?

A	$\frac{\pi}{4}$ rad	**B**	$\frac{\pi}{2}$ rad	**C**	$\frac{2\pi}{5}$ rad	**D**	$\frac{4\pi}{5}$ rad

Answer: B

$$\lambda = \frac{v}{f}$$
$$= \frac{320}{400}$$
$$= 0.8 \text{ m}$$

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

	$= 2\pi\left(\frac{0.2}{0.8}\right)$ $= \frac{\pi}{2}\,\text{rad}$